

StockSense – Technical Learning Notes

Module: Hardware Core Concepts & Signal Processing

Trainee: Kavindu Hasaranga | Role: Electronics & IoT Trainee | Date: October 2026

1. Load Cells, Strain Gauges & Wheatstone Bridge Dynamics

Strain Gauge Operating Principle

A strain gauge consists of an insulated flexible backing supporting a metallic foil pattern. When subject to mechanical force or weight, physical deformation alters its internal resistance (ΔR):

- **Tension (Stretching):** Gauge wire lengthens and thins → Electrical resistance (R) increases.
- **Compression (Squeezing):** Gauge wire shortens and thickens → Electrical resistance (R) decreases.

Wheatstone Bridge Configuration

Because fractional changes in resistance ($\Delta R/R$) are extremely minute, four strain gauges are configured in a Wheatstone bridge circuit to convert differential resistance shifts into a readable millivolt output (V_{out}):

$$V_{out} = V_E \cdot \left(\frac{R_1}{R_1 + R_2} - \frac{R_4}{R_3 + R_4} \right) \quad (1)$$

Where:

- V_E is the excitation voltage applied to the bridge (typically 5V or 3.3V).
- R_1, R_2, R_3, R_4 represent the four active strain gauge resistors on the load cell body.
- When no load is applied, the bridge is balanced ($V_{out} = 0$ mV). Load application unbalances the bridge, outputting a differential voltage proportional to weight.

2. The HX711 24-Bit ADC Module

Microcontrollers such as the ESP32 feature onboard 12-bit ADCs that lack the resolution and low-noise front-end needed to measure millivolt-level differential signals accurately. The HX711 chip provides two key hardware functions:

1. **Programmable Gain Amplifier (PGA):** Amplifies tiny millivolt inputs from Channel A by a selectable hardware gain of **128** or **64**.
2. **24-Bit Sigma-Delta ADC:** Digitizes the amplified analog signal into a 24-bit unsigned/signed integer representation:

$$\text{Resolution Range} = 0 \text{ to } (2^{24} - 1) = 0 \text{ to } 16,777,215 \text{ counts}$$

Hardware Interface & Control Lines

- **Bridge Terminals:** $E+$, $E-$ (Excitation Voltage Output), $A+$, $A-$ (Differential Analog Input).
- **Microcontroller Interface:**
 - PD_SCK (Power Down & Serial Clock): Clock pulses generated by the ESP32 GPIO.
 - DOUT (Data Out): Pulled LOW when data is ready; bits are shifted out on PD_SCK clock edges.

3. Mathematical Formulas for Weight & Quantity Derivation

Firmware Weight Calibration Equation

To convert raw 24-bit ADC counts into a physical calibrated mass (grams or kilograms), the edge firmware executes:

$$\text{Measured Weight} = \frac{\text{Raw ADC Read Value} - \text{Tare Offset}}{\text{Scale Calibration Factor}} \quad (2)$$

Where:

- **Tare Offset:** The zero-load ADC baseline reading (compensates for empty bin platform weight).
- **Scale Calibration Factor:** Digital counts per unit weight, derived during device calibration using a known test mass.

Discrete Item Quantity Calculation

Once total net weight is determined, item count is calculated in firmware or backend analytics as:

$$\text{Item Quantity} = \text{round} \left(\frac{\text{Measured Weight}}{\text{Unit Weight of Single Item}} \right) \quad (3)$$

Where:

- Unit Weight of Single Item is stored in the database for the active RFID/QR tag linked to that bin.

4. Next Study Topics (Upcoming Week 2 Prep)

- **RFID Identification Protocols:** SPI and I²C interface timing for MFRC522 and PN532 RFID readers.
- **MQTT Architecture & Transport:** Design of topic trees (`stocksense/site_id/bin_id/telemetry`), QoS 0 vs QoS 1 guarantees, and JSON payload definitions.